# 1.1 Evaluating Algebraic Expressions

Name _____ Period ____ Date _____

Algebra I 1.1 · Student sheet

**I can…**

- evaluate expressions with one variable.
- evaluate expressions with two or more variables using the order of operations.
- write and evaluate expressions for real-life situations.

## Part 1 · Order of operations relay

Evaluate each expression when **x = 3** and **y = −2**. Take turns: one partner does a step, the other checks it and does the next.

| Expression | Work | Value |
|---|---|---|
| $4x - 5$ | | |
| $x^2 + 2y$ | | |
| $3(x - y)$ | | |
| $(x + y)^2$ | | |
| $-y^2 + x$ | | |
| $xy - \frac{x}{3}$ | | |

## Part 2 · Notes: fill in the blanks

Order of operations: _____, _____, multiply and _____ (left to right), add and _____ (left to right).

To evaluate, _____ each variable with its value, using _____ around negatives.

$-3^2$ = _____, but $(-3)^2$ = _____.

## Part 3 · Examples (work them with the class)

### Example 1

Evaluate each expression.

a. $5x - 8$ when $x = 4$ b. $2x^2 + 1$ when $x = -3$

### Example 2

Evaluate $a^2 - 3b$ when $a = 4$ and $b = -2$.

### Example 3

Evaluate $\frac{x + y}{z} - 2y$ when $x = 7$, $y = 5$, and $z = 3$.

### Example 4

A phone plan costs $25 a month plus $0.10 per text. The cost is $C = 25 + 0.1t$, where $t$ is the number of texts. Find the cost for 140 texts.

## Part 4 · Partner practice

### Partner 1

Evaluate $3x^2 - x$ when $x = -2$.

### Partner 2

Evaluate $4m - n^2$ when $m = 3$ and $n = -5$.